

Introducing smalltalk-dev-plugin

AI-Driven Development Toolkit for Pharo Smalltalk

Masashi Umezawa

<https://github.com/mumez/smalltalk-dev-plugin>

What is smalltalk-dev-plugin?

A comprehensive toolkit that bridges **AI code editors** and the **Pharo Smalltalk** environment.

Using AI coding agents with Smalltalk doesn't work well out of the box:

- **No knowledge of Pharo** — AI doesn't know how to interact with the environment
- **Unknown project structure** — Typical Tonel/package layout is unfamiliar
- **No testing/debugging skills** — Can't run tests or diagnose errors
- **No library access** — Can't browse existing classes or methods
- **Poor coding style** — Doesn't know Smalltalk idioms and conventions

This is a major barrier for developers who want AI assistance in Smalltalk.

We gave AI the complete skillset for Smalltalk development:

Challenge	Solution
Environment interaction	MCP tools for Pharo communication
Project structure	<code>/st-setup-project</code> with templates
Testing & debugging	<code>/st-test</code> , <code>/st-eval</code> , debugger skill
Library navigation	Finder skills, code search tools
Code quality	Validator, linter, commenter agent

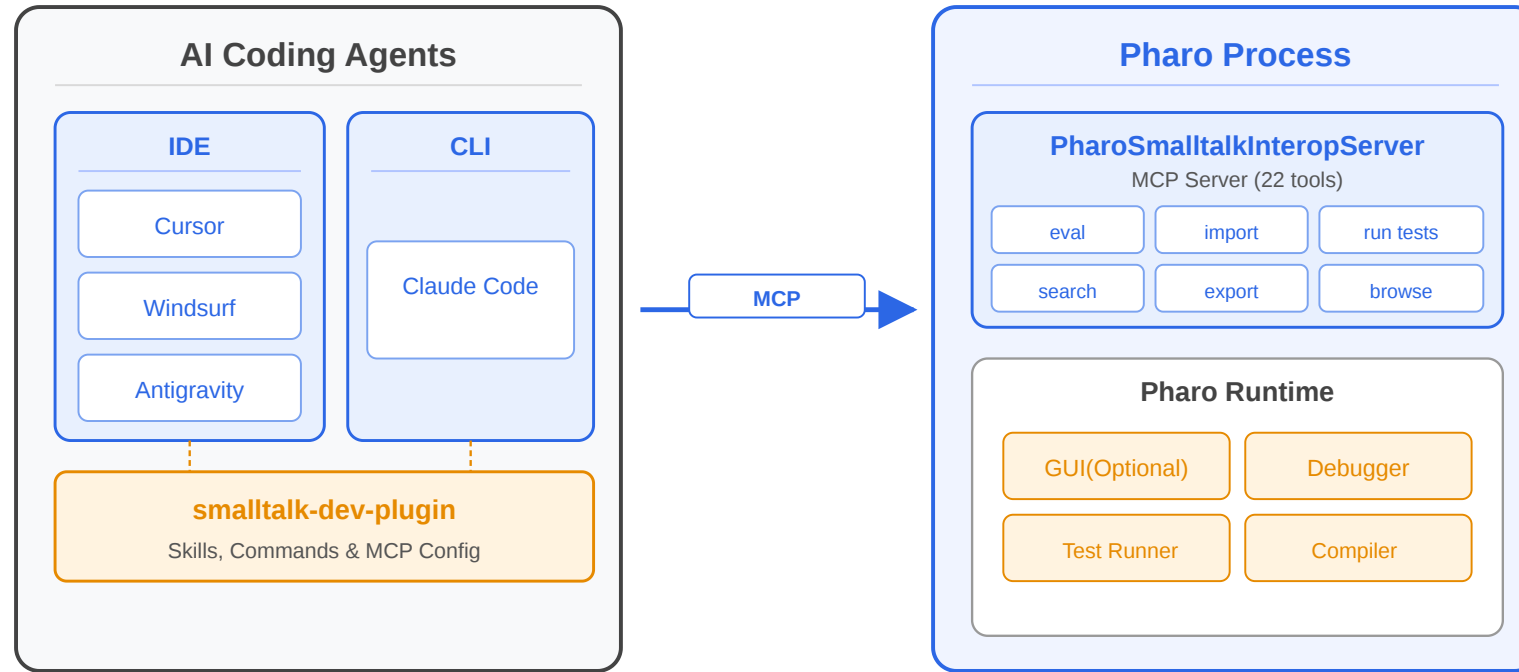
Result: AI becomes a reliable Smalltalk development partner!

"AI editing as the source of truth"

| *Edit Tonel files in the AI editor, import to Pharo, test, and iterate.*

At a Glance

- 9 slash commands
- 5 specialized AI skills
- 22 MCP tools for Pharo interaction
- Automated code quality checks and documentation generation



- Existing AI agents talk to Pharo via MCP — no custom IDE needed
- Even works with headless Pharo — usable in CI and remote agents like Devin

Agent	Support Level
Claude Code	Full support (primary)
Cursor	Simplified setup via script
Windsurf	Simplified setup via script
Antigravity	Simplified setup via script
GitHub Copilot CLI	Simplified setup via script
OpenCode	Simplified setup via script

The rest of this presentation focuses on the Claude Code applied version.

Installation

Installation (1) — Install the Plugin

```
# Add marketplace from GitHub
claude plugin marketplace add mumez/smalltalk-dev-plugin

# Install the plugin
claude plugin install smalltalk-dev
```

Option A: Docker (Easy)

Use [smalltalk-interop-docker](#) to run a pre-configured Pharo image:

```
docker compose up -d
```

Option B: Local Pharo

Install and start the interop server manually:

```
Metacello new
  baseline: 'PharoSmalltalkInteropServer';
  repository: 'github://mumez/PharoSmalltalkInteropServer:main/src';
  load.

SisServer current start.
```

Setup scripts are provided for each agent:

```
./extra/setup-cursor.sh [target-directory]
./extra/setup-windsurf.sh [target-directory]
./extra/setup-antigravity.sh [target-directory]
./extra/setup-copilot.sh [target-directory] # GitHub Copilot CLI
./extra/setup-opencode.sh [target-directory] # OpenCode
```

Limitations

- Some features may differ depending on the agent
- Pharo-side setup is the same as for Claude Code

Basic Usage

Run `claude` in an **empty directory** or your **existing project directory**.

- You can also point to an Iceberg repository directory
- However, to avoid conflicts, it is better to **clone the repo into a separate directory** and use that instead

`/st-buddy` is a friendly assistant that routes to the right tool.

Just describe what you want in natural language:

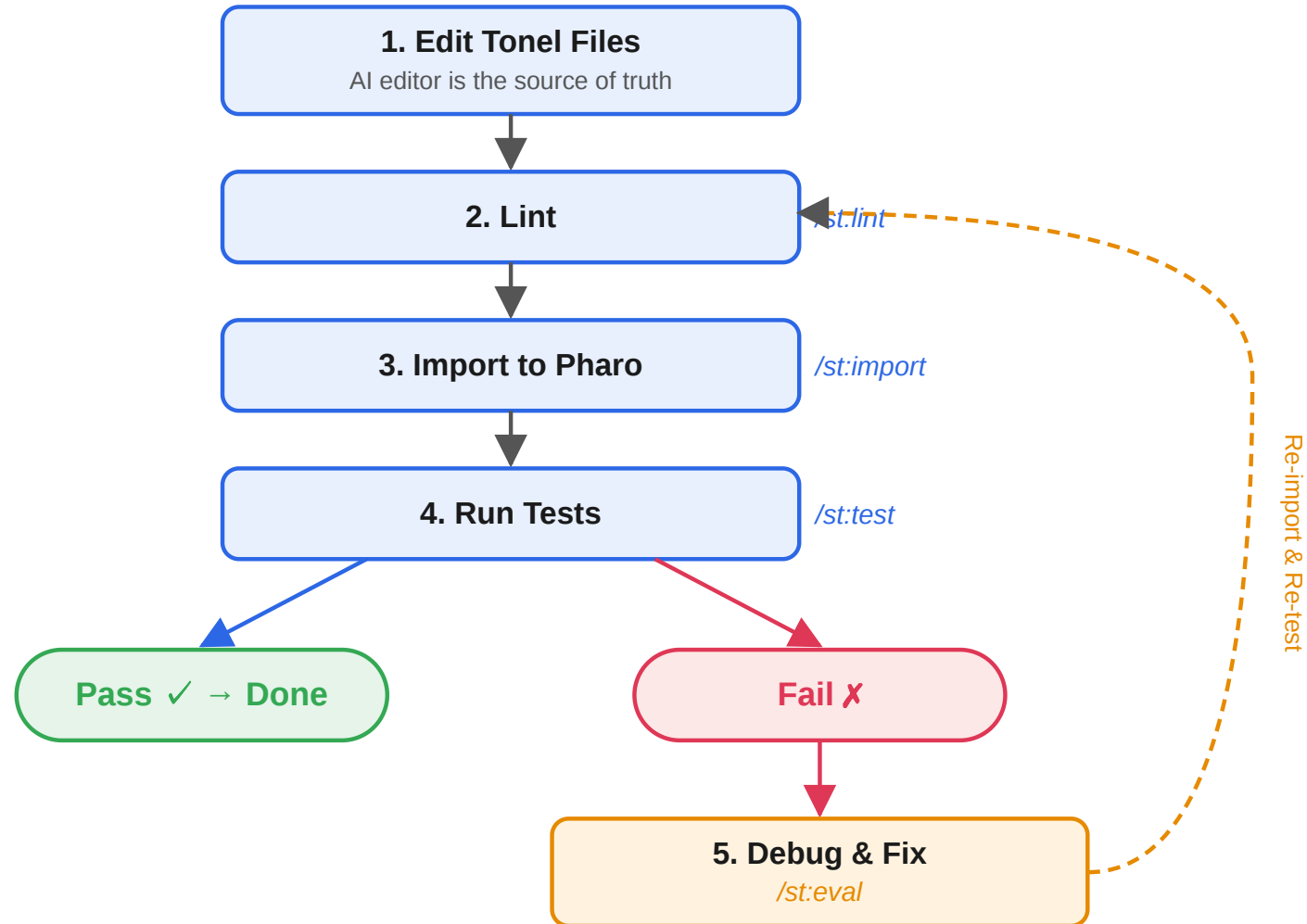
```
/st-buddy I want to create a Person class with name and age
```

The assistant will:

1. Load the appropriate skill automatically
2. Create the Tonel files
3. Guide you through linting, importing, and testing

Your Intent	Loaded Skill
"Create...", "Add..."	smalltalk-developer
"Error...", "Test failed..."	smalltalk-debugger
"How do I use...?"	smalltalk-usage-finder
"Who implements...?"	smalltalk-implementation-finder

You don't need to remember which tool to use — just talk to `/st-buddy`.



Components

smalltalk-developer

The core development skill. Knows Tonel format, package structure, and the full Edit → Lint → Import → Test workflow.

smalltalk-debugger

Systematic debugging specialist. Diagnoses errors, runs incremental code execution with `/st-eval`, and guides you to the fix.

smalltalk-usage-finder

Discovers how classes and methods are used. Finds examples, analyzes usage patterns, and helps you understand unfamiliar APIs.

smalltalk-implementation-finder

Finds method implementations across class hierarchies. Useful for learning coding idioms and assessing refactoring impact.

smalltalk-commenter

Automatically suggests **CRC-style class comments** for undocumented classes. Triggered after file edits via hooks — non-intrusive and prioritizes classes that need documentation most.

Command	Purpose
<code>/st-buddy</code>	Friendly assistant (main entry)
<code>/st-init</code>	Initialize development session
<code>/st-setup-project</code>	Create project boilerplate
<code>/st-eval</code>	Execute Smalltalk expressions
<code>/st-import</code>	Import Tonel packages to Pharo
<code>/st-export</code>	Export packages from Pharo
<code>/st-test</code>	Run SUnit tests
<code>/st-lint</code>	Check code quality
<code>/st-validate</code>	Validate Tonel syntax

Two MCP servers power the integration behind the scenes:

pharo-interop (22 tools)

Pharo communication — eval, import/export, tests, code navigation, etc.

smalltalk-validator (5 tools)

Tonel validation and linting.

Note: You rarely need to interact with these directly.

`/st-buddy` handles orchestration for you.

Live Examples

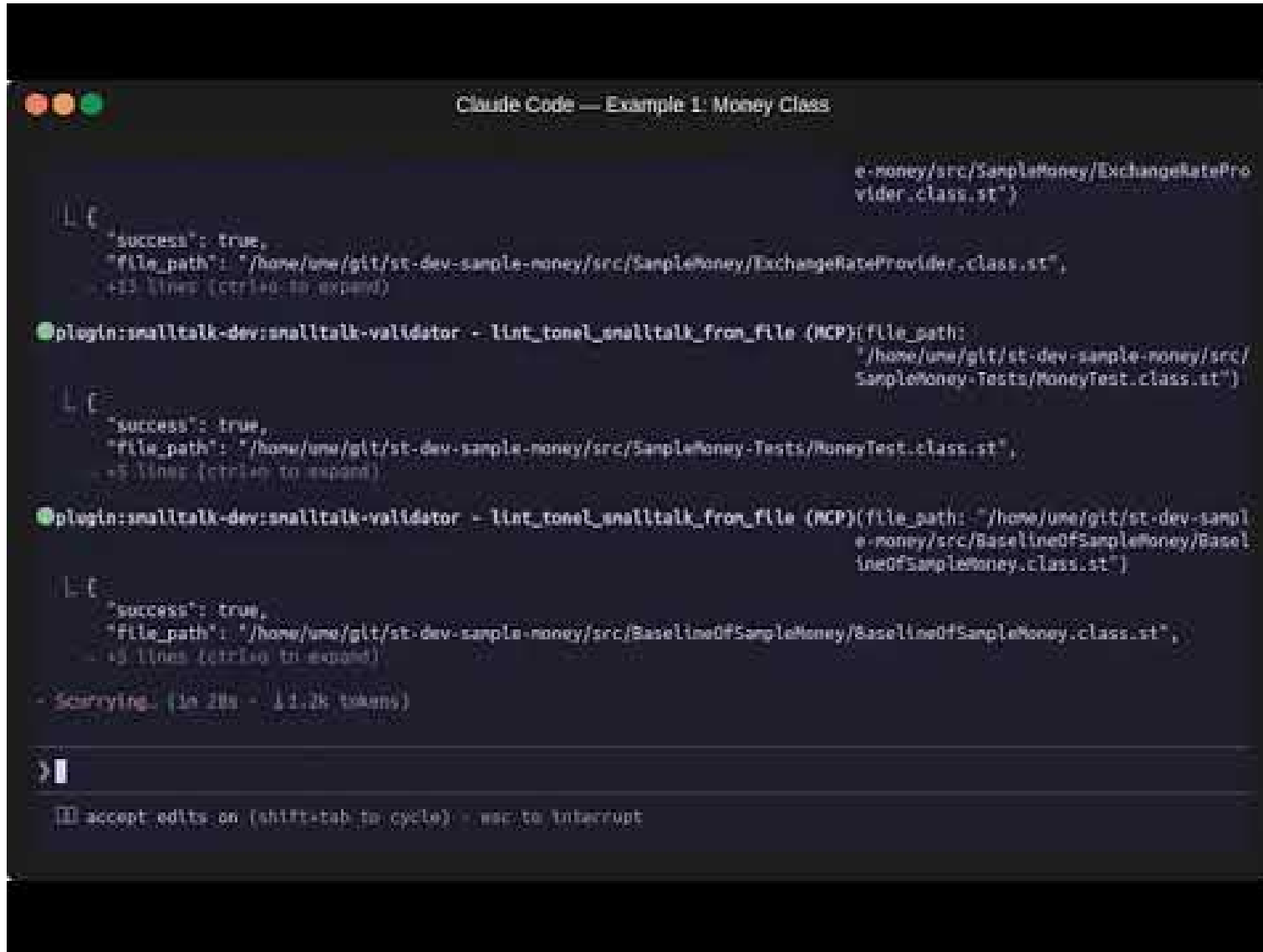
Example 1 — Starting from Scratch

A simple use case: starting from an empty directory as a Smalltalk beginner.

```
/st-buddy I'm a Smalltalk beginner. I want to create a Money class  
that can perform arithmetic operations across different currency units.  
Please help me from the project initialization.
```


Example 1 — What Happens Next

25



```
Claude Code — Example 1: Money Class

e-money/src/SampleMoney/ExchangeRateProvider.class.st")

L E
{"success": true,
 "file_path": "/home/una/git/st-dev-sample-money/src/SampleMoney/ExchangeRateProvider.class.st",
 +11 lines (ctrl+q to expand)

plugin:smalltalk-dev:smalltalk-validator - lint_tonel_smalltalk_from_file (MCP)(file_path:
"/home/una/git/st-dev-sample-money/src/
SampleMoney-Tests/MoneyTest.class.st")

L E
{"success": true,
 "file_path": "/home/una/git/st-dev-sample-money/src/SampleMoney-Tests/MoneyTest.class.st",
 +5 lines (ctrl+q to expand)

plugin:smalltalk-dev:smalltalk-validator - lint_tonel_smalltalk_from_file (MCP)(file_path: "/home/una/git/st-dev-sampl
e-money/src/BaselineOfSampleMoney/BaselineOfSampleMoney.class.st")

L E
{"success": true,
 "file_path": "/home/una/git/st-dev-sample-money/src/BaselineOfSampleMoney/BaselineOfSampleMoney.class.st",
 +9 lines (ctrl+q to expand)

- Scarrying... (in 28s - 11.2k tokens)

> |
[ ] accept edits on (shift+tab to cycle) - esc to interrupt
```

- [Video](#)
- [Generated source](#)
- [Demo](#)

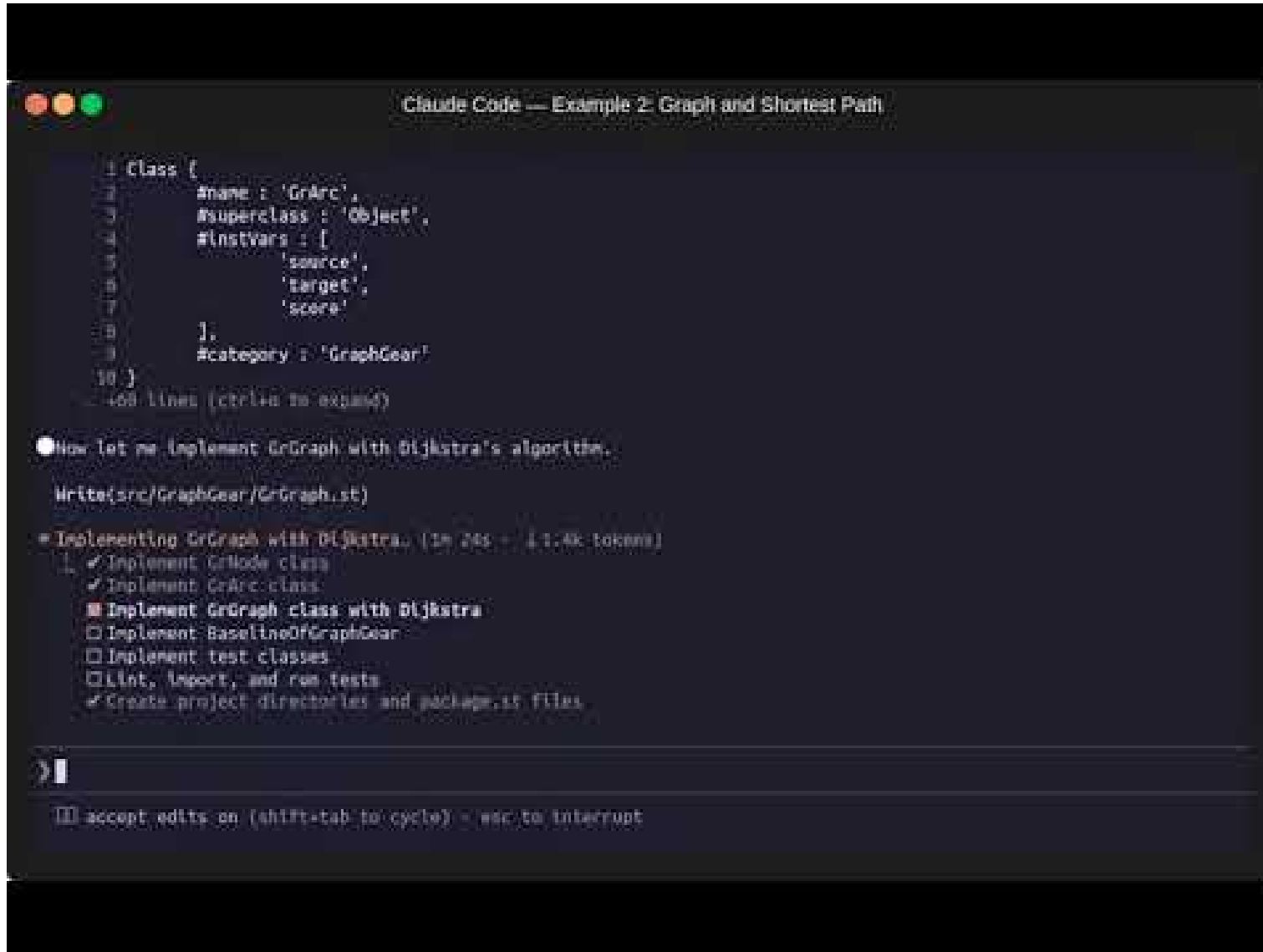
Example 2 — Graph Algorithms

A more complex use case for someone familiar with Smalltalk.

```
/st-buddy I want to create GrNode and GrArc to represent directed graphs  
and solve shortest path problems. Nodes have a name, arcs have a score.  
Let's start this as a GraphGear project.
```

Example 2 — What Happens Next

27



```
Claude Code — Example 2: Graph and Shortest Path

1 class {
2   #name : 'GrArc',
3   #superclass : 'Object',
4   #instvars : [
5     'source',
6     'target',
7     'score'
8   ],
9   #category : 'GraphGear'
10 }
11 }
12 }
13 }
14 }
15 }
16 }
17 }
18 }
19 }
20 }
21 }
22 }
23 }
24 }
25 }
26 }
27 }
28 }
29 }
30 }
31 }
32 }
33 }
34 }
35 }
36 }
37 }
38 }
39 }
40 }
41 }
42 }
43 }
44 }
45 }
46 }
47 }
48 }
49 }
50 }
51 }
52 }
53 }
54 }
55 }
56 }
57 }
58 }
59 }
60 }
61 }
62 }
63 }
64 }
65 }
66 }
67 }
68 }
69 }
70 }
71 }
72 }
73 }
74 }
75 }
76 }
77 }
78 }
79 }
80 }
81 }
82 }
83 }
84 }
85 }
86 }
87 }
88 }
89 }
90 }
91 }
92 }
93 }
94 }
95 }
96 }
97 }
98 }
99 }
100 }
```

468 lines (ctrl+u to expand)

Now let me implement GrGraph with Dijkstra's algorithm.

Write(src/GraphGear/GrGraph.st)

Implementing GrGraph with Dijkstra. (1m 24s - 11.4k tokens)

- ✓ Implement GrNode class
- ✓ Implement GrArc class
- Implement GrGraph class with Dijkstra
- Implement BaselineOfGraphGear
- Implement test classes
- Lint, import, and run tests
- ✓ Create project directories and package.st files

> |

accept edits on (shift+tab to cycle) - esc to interrupt

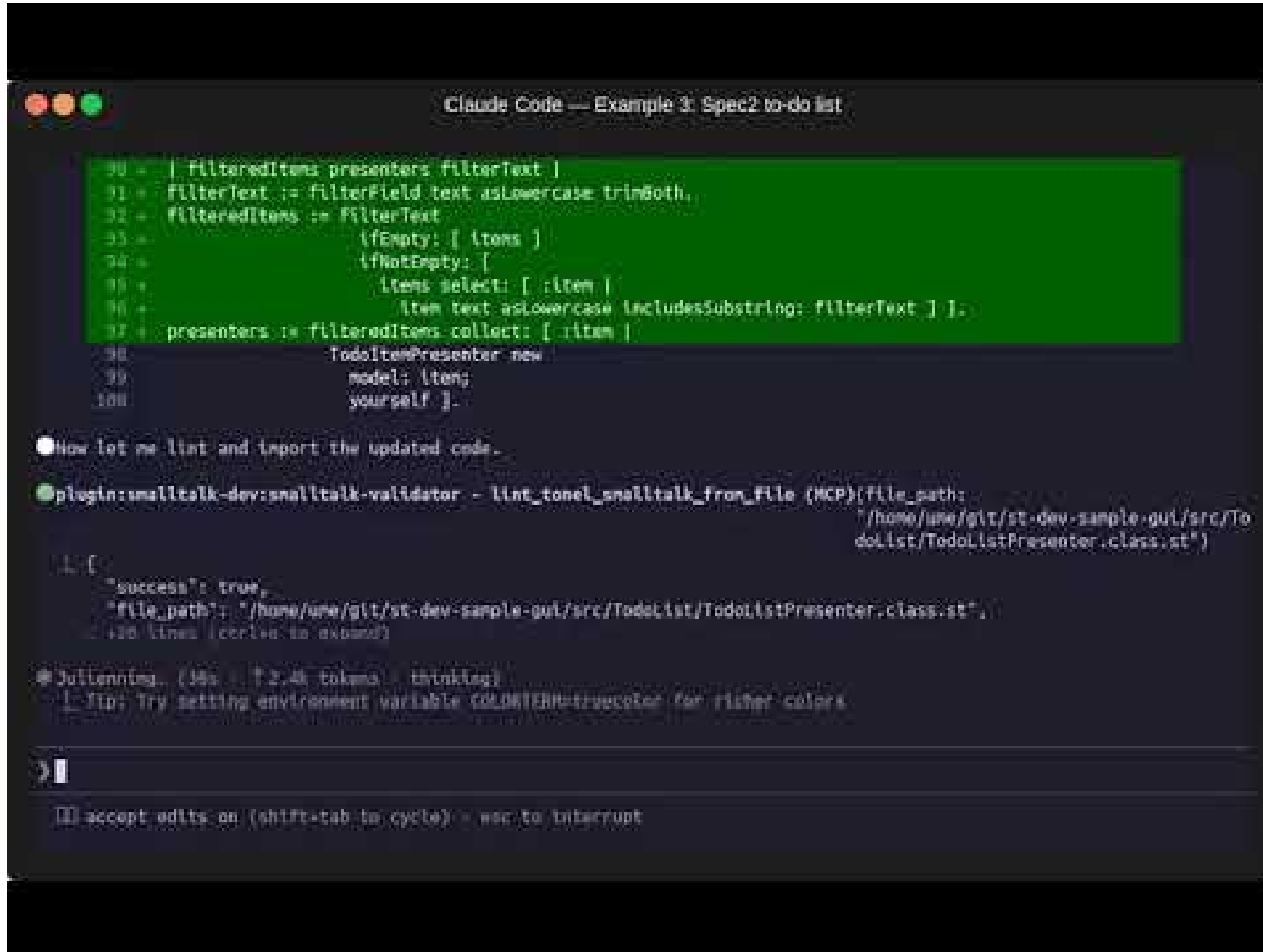
- [Video](#)
- [Generated source](#)
- [Demo](#)

Building a GUI application with Spec2.

```
/st-buddy I want to make a simple to-do list using the Spec2 framework.  
Include a checkbox for each item, plus an input field  
with Add/Remove buttons at the bottom.  
Only checked items can be removed. Start developing.
```

Example 3 — What Happens Next

29



The screenshot shows a Claude Code window titled "Claude Code — Example 3: Spec2 to-do list". The code editor displays a Smalltalk snippet with line numbers 90 to 100. The code defines a filter for a to-do list and creates a presenter. Below the code, a JSON response is shown, indicating a successful lint operation. The response includes a success flag, the file path, and a line count. A tip from Julienning is also visible at the bottom.

```
90 + | filteredItems presenters filterText |
91 + filterText := filterField text asLowercase trimBoth.
92 + filteredItems := filterText
93 +     ifEmpty: [ items ]
94 +     ifNotEmpty: [
95 +         items select: [ :item |
96 +             item text asLowercase includesSubstring: filterText ] ].
97 + presenters := filteredItems collect: [ :item |
98 +     TodoItemPresenter new
99 +         model: item;
100 +         yourself ].
```

```
{
  "success": true,
  "file_path": "/home/une/git/st-dev-sample-gui/src/TodoList/TodoListPresenter.class.st",
  "lines": 130
}
```

Tip: Try setting environment variable COLORTERM=truecolor for richer colors

- [Video](#)
- [Generated source](#)
- [Demo](#)

Case Studies

Case Studies — Real Projects

Projects created using smalltalk-dev-plugin.

Both were **over 90% built with Claude Code + smalltalk-dev-plugin.**

A library for connecting to AI agents from the Pharo side using the ACP protocol.

Connects to: Gemini CLI, Claude Code, OpenCode, Copilot CLI, Codex CLI, etc.

<https://github.com/mumetz/pharo-acp>

A simple AI Chat GUI for Pharo. Uses pharo-acp internally.

<https://github.com/mumetz/pharo-acp-chat-ui>



smalltalk-dev-plugin makes Pharo Smalltalk development with AI assistants practical and productive.

Just type `/st-buddy` and start building.

Feedback and contributions are welcome!

<https://github.com/mumez/smalltalk-dev-plugin>